

# Integer Occurrences in Rational Linear Recurrences

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# The Big Question

## Integer Coefficients

$$S_0 = 1, S_1 = 7$$

$$S_{n+2} = 3S_{n+1} + 2S_n$$

Clearly all terms are integers.

## Rational Coefficients?

$$S_0 = 1, S_1 = 1$$

$$S_{n+2} = \frac{1}{3}S_{n+1} + \frac{1}{2}S_n$$

Can we get integers?

**When do integers appear in rational recurrences?**

# Our Setting

## Rational Second-Order Linear Recurrence

Given  $S_0, S_1 \in \mathbb{Z}$ , define

$$S_{n+2} = c_1 S_{n+1} + c_0 S_n$$

where  $c_1 = \frac{c}{d}$  and  $c_0 = \frac{a}{b}$  are **irreducible** fractions.

**We study:**  $b > 1$  or  $d > 1$  (at least one rational coefficient)

When  
integers  
appear?

Consecutive  
integers?

Isolated  
integers?

Finitely  
many?

## Example 1: Simple Case

$$S_{n+2} = \frac{1}{3}S_{n+1} + \frac{1}{2}S_n \text{ with } S_0 = 2, S_1 = 3$$

| $n$ | $S_n$             | Integer? |
|-----|-------------------|----------|
| 0   | 2                 | YES      |
| 1   | 3                 | YES      |
| 2   | 2                 | YES      |
| 3   | $\frac{13}{6}$    | NO       |
| 4   | $\frac{31}{18}$   | NO       |
| 5   | $\frac{121}{108}$ | NO       |

**Three consecutive integers!** Then no more.

# The Key Transformation

**Main Idea:** Convert to an integer sequence to study divisibility.

Let  $Q = \text{lcm}(b, d)$ . Define the **auxiliary sequence**:

$$T_n := Q^n S_n$$

## Key Properties

- $(T_n)$  satisfies an **integer** recurrence:  $T_{n+2} = PT_{n+1} + RT_n$
- Where  $P = c \cdot \frac{Q}{d}$  and  $R = a \cdot Q \cdot \frac{Q}{b}$  are both integers
- **Critical:**  $S_n \in \mathbb{Z} \Leftrightarrow Q^n \mid T_n$

Integrality problem  $\rightarrow$  Divisibility problem

## Example 2: Auxiliary Sequence in Action

For  $S_{n+2} = \frac{1}{3}S_{n+1} + \frac{1}{2}S_n$ :  $Q = 6$ ,  $P = 2$ ,  $R = 18$

Auxiliary recurrence:  $T_{n+2} = 2T_{n+1} + 18T_n$  with  $T_0 = 2$ ,  $T_1 = 18$

| $n$ | $T_n$ | $Q^n = 6^n$ | $Q^n \mid T_n?$ | $S_n = T_n/Q^n$ |
|-----|-------|-------------|-----------------|-----------------|
| 0   | 2     | 1           | YES             | 2               |
| 1   | 18    | 6           | YES             | 3               |
| 2   | 72    | 36          | YES             | 2               |
| 3   | 468   | 216         | NO              | $\frac{13}{6}$  |
| 4   | 4104  | 1296        | NO              | $\frac{57}{18}$ |

Check:  $72 = 36 \cdot 2$      $468 = 216 \cdot \frac{13}{6}$

# $p$ -adic Valuations

**Tool:** Use  $p$ -adic valuations  $\nu_p(x)$  = highest power of  $p$  dividing  $x$

## $p$ -adic Integrality Criterion

$$S_n \in \mathbb{Z} \Leftrightarrow \nu_p(T_n) \geq n\nu_p(Q) \text{ for every prime } p \mid Q$$

Example:  $Q = 6 = 2 \cdot 3$

Need **both**:

- $\nu_2(T_n) \geq n$  (enough factors of 2)
- $\nu_3(T_n) \geq n$  (enough factors of 3)

This allows us to analyze each prime separately!

# Connection to Lucas Sequences

The auxiliary sequence connects to **Lucas sequences**:

## Lucas Sequence

$$\begin{cases} U_0 = 0 \\ U_1 = 1 \\ U_{n+2} = PU_{n+1} + RU_n \end{cases}$$

## Companion Identity

$$T_n = T_1 U_n + RT_0 U_{n-1}$$

**Why this is powerful:**

- Lucas sequences have well-studied  $p$ -adic properties
- Results from Ballot [1] give exact formulas for  $\nu_p(U_n)$
- We can use these to understand when  $T_n$  has enough divisibility



# Finding Long Runs of Integers

## Theorem (Constructive Initial Integrality)

For any  $m \geq 1$  and any rational coefficients, we can find  $S_0, S_1 \in \mathbb{Z}$  such that

$S_0, S_1, S_2, \dots, S_m$  are **all** integers

**How?** Choose initial conditions with **very high**  $p$ -adic valuations:

$$S_0 = \prod_{p|Q} p^{A_{0,p}(m)}, \quad S_1 = \frac{1}{Q} \prod_{p|Q} p^{A_{1,p}(m)}$$

where the exponents are carefully computed from Lucas sequence valuations.

**Key insight:** Start with enough factors of each prime!

## Example 3: Computational Results

$$S_{n+2} = \frac{23}{2}S_{n+1} + 70S_n \text{ with carefully chosen } S_0, S_1$$

Powers of 2 dividing  $S_n$  (i.e.,  $\nu_2(S_n)$ ) for different initial conditions:

| Initial conditions | $\nu_2$ pattern                                                    |
|--------------------|--------------------------------------------------------------------|
| (81, 76)           | 0, 2, 4, 6, 8, 10, 12, <b>15</b> , 13, 12, ..., 0<br>22 terms!     |
| (6247, -7084)      | 0, 2, 4, 6, 8, 10, 12, 14, 16, <b>16</b> , 15, ..., 0<br>26 terms! |
| (20037, -20036)    | 0, 2, 4, 6, 8, ..., 18, <b>20</b> , 20, 19, ..., 0<br>32 terms!    |
| (301487, -247884)  | 0, 2, 4, 6, 8, ..., 20, <b>22</b> , 22, 21, ..., 0<br>35 terms!    |

*Pattern increases with larger initial conditions!*

# Visualizing the Pattern

$$S_{n+2} = \frac{23}{2}S_{n+1} + 70S_n \text{ with } (81, 76)$$

**Binary representation of terms:**

$$S_0 = 81 = 1010001_2$$

$$S_1 = 76 = 1001100_2$$

$$S_2 = 6544 = 1100110010000_2$$

$$S_3 = 80576 = 10011101011000000_2$$

$$S_4 = 1384704 = 101010010000100000000_2$$

$\vdots$

Notice the trailing zeros increase (these are the powers of 2)!

The pattern: grows, peaks at  $n = 7$ , then decreases back to 0 at  $n = 21$ .

# Combining Solutions: The Gluing Method

**Can we extend integer runs even further?**

**Idea:** If we have two sequences with runs of length  $m_0$  and  $m_1$ , combine:

$$\text{New sequence} = u \cdot (\text{Sequence 0}) + v \cdot (\text{Sequence 1})$$

for carefully chosen integers  $u, v$ .

## Example

Starting from  $(11, 4)$  with 11 integer terms and  $(5, -4)$  with 8 terms:

$$8 \cdot (11, 4) + (11, 4) = (51, -28)$$

gives a sequence with **13 integer terms!**

This allows iterative construction of longer and longer runs.

# Restrictions on Consecutive Integers

Not everything is possible!

## Theorem (No Consecutive Integers)

If  $\gcd(b, d) = 1$ ,  $b > 1$ ,  $d > 1$ , and  $\gcd(S_0, S_1) = 1$ , then:

$S_n$  and  $S_{n+1}$  **cannot both** be integers for any  $n \geq 2$

**Intuition:** The coprimality forces divisibility constraints that propagate backward. If two consecutive terms far out are integers, we can show  $S_0$  and  $S_1$  must share a common factor, contradicting  $\gcd(S_0, S_1) = 1$ .

## Example 4: No Consecutive Integers

$$S_{n+2} = \frac{1}{3}S_{n+1} + \frac{1}{2}S_n \text{ with } S_0 = 1, S_1 = 1$$

Here  $b = 2, d = 3$ , so  $\gcd(b, d) = 1$  and  $\gcd(1, 1) = 1$

| $n$ | $S_n$           | Status      |
|-----|-----------------|-------------|
| 0   | 1               | Integer     |
| 1   | 1               | Integer     |
| 2   | $\frac{5}{6}$   | Not integer |
| 3   | $\frac{7}{9}$   | Not integer |
| 4   | $\frac{52}{81}$ | Not integer |

**Confirmed:** No consecutive integers for  $n \geq 2$

## Definition

A prime  $p$  is **special** for  $(P, R)$  if both  $p \mid P$  and  $p \mid R$ .

When  $p$  is special,  $\nu_p(U_n)$  has predictable structure (from Ballot [1]):

Let  $a = \nu_p(P)$ ,  $b = \nu_p(R)$ . If  $b < 2a$ :

$$\nu_p(U_{2n+1}) = bn$$

$$\nu_p(U_{2n}) = bn + (a - b) + \nu_p(n) + \lambda_n$$

**Key observation:** The valuation at **even** indices grows faster!

This creates opportunities for isolated integers at specific positions.

## Theorem (Prescribed Integers)

Under suitable conditions on the primes, we can construct  $S_0, S_1$  such that

$$S_{2m_1}, S_{2m_2}, \dots, S_{2m_k} \text{ are integers}$$

for **any** specified even indices  $2m_1 < 2m_2 < \dots < 2m_k$ !

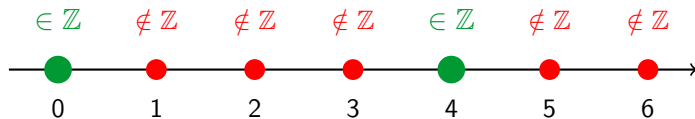
## Method:

- 1 Each index  $m_j$  gives a congruence condition on  $S_0/S_1 \pmod{p^{L(m_j)}}$
- 2 Space indices appropriately so congruences are compatible
- 3 Use Chinese Remainder Theorem to satisfy all conditions simultaneously



## Example 5: Integer Islands

**Conceptual example:** Create integers only at  $n = 0$  and  $n = 4$



Even with  $\gcd(S_0, S_1) = 1$ , we can create isolated integers at prescribed locations using  $p$ -adic congruences!

# Only Finitely Many Integers

For a **fixed** sequence, integers can't keep appearing forever.

## Theorem (Finiteness)

Fix the recurrence and initial conditions  $S_0, S_1$ . If the characteristic roots satisfy  $r_1/r_2$  not a root of unity, then there exists explicit  $N$  such that

$$S_n \notin \mathbb{Z} \text{ for all } n > N$$

**Why?** Using Binet formula  $S_n = \alpha r_1^n + \beta r_2^n$ :

- Integrality requires  $\nu_p(\Gamma \Lambda^n + 1) \geq np - C$  (linear growth in  $n$ )
- But  $p$ -adic linear forms theorem (Chim [2]) gives upper bound  $\leq K(\log n + C')$  (logarithmic!)
- Linear eventually beats logarithmic  $\Rightarrow$  explicit cutoff

## Example 6: Explicit Cutoff

$$S_{n+2} = \frac{5}{2}S_{n+1} + \frac{3}{4}S_n \text{ with } S_0 = 1, S_1 = 1$$

Characteristic roots:  $r_1 = 3$ ,  $r_2 = -\frac{1}{4}$

Since  $Q = 4 = 2^2$  and  $\nu_2(r_2) = -2$ , we need  $\nu_2(S_n) \geq 0$ , which forces

$$\nu_2(\Gamma\Lambda^n + 1) \geq 2n - \text{constant}$$

But the  $p$ -adic logarithm bound gives:

$$\nu_2(\Gamma\Lambda^n + 1) \leq K_2(\log n + C_H)$$

Solving  $2n - C_1 \leq K_2(\log n + C_H)$  gives **explicit**  $N$ .

**Conclusion:** Only finitely many integers in this sequence!

## What We Proved About Integer Occurrences

**Construction:** Can create arbitrarily long runs of consecutive integers at the start (choose initial conditions with high  $p$ -adic valuations)

**Restriction:** With coprime denominators and  $\gcd(S_0, S_1) = 1$ , no two consecutive terms beyond  $S_2$  can both be integers

**Islands:** Under special prime conditions, can prescribe isolated integers at arbitrary even indices

**Finiteness:** For any fixed sequence, only finitely many integers appear (with explicit computable bound)

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Lower bounds for linear forms in two  $p$ -adic logarithms.  
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**Thank you!**

Questions?